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Insight into keyhole and melt pool dynamics in laser welding of zinc-coated steels by means of high-speed synchrotron X-ray imaging

Klaus Schricker^{a,*}, Peter Hellwig^a, Leander Schmidt^a, Christian Diegel^a, Marc Seibold^a, Hannes Friedmann^a, Fabian Fröhlich^a, Falk Nagel^b, Peter Kallage^b, Yunhui Chen^{c,d}, Herwig Requardt^d, Alexander Rack^d, Jean Pierre Bergmann^a

^aTechnische Universität Ilmenau, Production Technology Group, Ilmenau, Germany

^bCOHERENT, Hamburg, Germany

^cRMIT University, School of Engineering, Melbourne, Australia

^dESRF – The European Synchrotron, Grenoble, France

* Corresponding author. Tel.: +49-3677-69-3808; fax: +49-3677-69-1660. E-mail address: klaus.schricker@tu-ilmenau.de

Abstract

Laser welding zinc-coated steels is of major importance in automotive engineering and other industries to ensure cost-effective corrosion resistance of assemblies without requiring post-weld rework. However, the low evaporation temperature of zinc is causing welding defects, in particular melt ejections and spatter. In this work, in-situ experiments of laser beam welding were performed using high-speed synchrotron X-ray imaging (up to 40,000 images/second) to determine the dynamics in the keyhole and in the weld pool and to provide detailed explanations for the formation of weld defects. A simplified sample geometry made it possible to weld zinc-coated steel sheets (DX51D Z275) in a lap joint (sheet thickness 1.25 mm each) with a fiber laser (COHERENT HighLight FL-ARM 8000) to describe fundamental phenomena. The high spatial and temporal resolution of the radiography allowed describing the acting mechanisms and their effect on keyhole and melt pool.

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1. Introduction

Laser welding of zinc-coated steels is important for various applications, particularly in automotive engineering, to achieve cost-effective corrosion protection. Lap joints are commonly used due to their high accessibility, low tolerance requirements, and easy-to-implement clamping technology.

The combination of galvanized steels and lap joints in welding presents challenges due to the low evaporation temperature of the zinc coating (approx. 906 °C) compared to the melting temperature of the steel (≥ 1250 °C). During laser beam welding, the high temperatures cause the zinc coating to evaporate accompanied by a high zinc vapor pressure [1]. This vapor pressure is crucial for welds in lap joints with zero-gap

because degassing takes place through the processing zone [2, 3]. On the one hand, it is stated that the zinc vapor can destabilize the keyhole that acts as degassing channel [1, 4-6]. On the other hand, the zinc vapor can interact with the melt pool. In this case, lateral degassing is a major issue, i.e., the heat-affected zinc layer adjacent to the melt pool widens, causing an increased evaporation of the zinc coating and therefore increasing vapor pressure interacting with the melt pool [7]. Both effects can lead to the expulsion of significant fractions of melt, resulting in major welding defects such as spatters and cavities [8]. The effect of zinc vapor pressure can be limited by introducing degassing gaps [9], but requires more effort compared to welding in zero-gap configuration.

High-speed imaging in half-section setups [1, 10, 11] and numerical simulations [11, 12] has been used to gain a time-dependent and in-depth understanding of phenomena within the process zone, whereby X-ray imaging was used for validating the keyhole geometry [11]. However, detailed descriptions of time-dependent phenomena inside the workpiece leading to different defects based on experimental studies have not been provided yet. High-speed radiography using synchrotron radiation has a significant potential for visualizing hidden processes at high recording rates [13] while limiting the effect of the setup on the resulting weld, e.g., compared to welding behind glass plates. For instance, research has shown that laser beam welding of copper at a rate of 20,000 images/second [14] and high-alloy steels at a rate of up to 100,000 images/second is possible [15]. In this paper, high-speed synchrotron X-ray imaging is utilized to describe transient processes in the keyhole and melt pool during laser beam welding of zinc-coated steels in the technical zero gap.

2. Materials and methods

2.1. Laser welding setup

The welding process was carried out with a COHERENT HighLight FL8000-ARM fiber laser with an adjustable ring mode (ARM) fiber consisting of core and concentric ring intensity. The focal diameters of core and ring intensity $d_{f,core}$ and $d_{f,ring}$ were determined by measurements with a Primes FocusMonitor based on the D86 beam width measurement method at a max. beam power of 600 W in total to avoid damaging the measurement device. The laser welding setup is summarized in Table 1. It should be noted that the systematic study of different intensity profiles was not in the focus of the experiments presented, i.e., the beam profile was adjusted to certain penetration depths. Specific information on laser parameters used is provided for each experiment discussed in section 3.

Table 1. Laser welding setup

Parameter	Symbol	Quantity and unit
Laser beam power in core	$P_{L,core}$	up to 4,000 W
Laser beam power in ring	$P_{L,ring}$	up to 4,000 W
Wavelength laser beam	λ_{Laser}	1,070 nm
Focal diameter of core	$d_{f,core}$	89 μ m
Focal diameter of ring	$d_{f,ring}$	208 μ m
Welding speed	v_w	6 m/min
Focal position	z_0	0 / -3 mm

2.2. High-speed synchrotron X-ray imaging setup

Fig. 1 provides a schematic of the in situ high-speed synchrotron X-ray imaging setup. The experiments were carried out at European Synchrotron Radiation Facility (ESRF) at beamline ID19. The X-ray beam size was $4 \times 4 \text{ mm}^2$ and the energy $\leq 60 \text{ keV}$. A Ce-doped $\text{Lu}_3\text{Al}_5\text{O}_{12}$ scintillator with a thickness of 250 μ m from manufacturer Crytur was lens-coupled (5x) to a Photron SA-Z high-speed camera propagation distance: 4.5 m, acquisition rate f_1 : 40,000 images/second at

$640 \times 760 \text{ pixel}^2$, acquisition rate f_2 : 20,000 images/second at $1024 \times 1024 \text{ pixel}^2$). A second high-speed camera Photron SA-X2 was equipped with a 630-40 nm bandpass filter and used in combination with a high-power LED ($\lambda_{LED} = 630 \text{ nm}$) to record the sheet surface simultaneously to the X-ray imaging (10,000 images/second at $1024 \times 400 \text{ pixel}^2$).

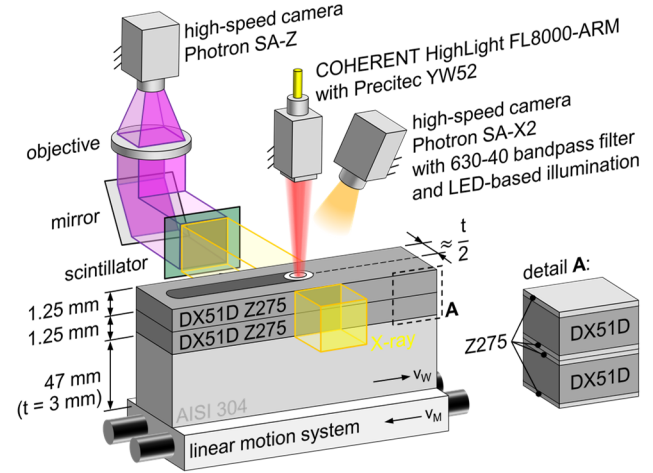


Fig. 1. High-speed synchrotron X-ray imaging setup

Zinc-coated steel sheets (DX51D Z275) were positioned in zero-gap lap configuration for the welding experiments. Both sheets were mounted on an AISI 304 substrate that acted as sample carrier. The sample thickness t in X-ray direction was 3 mm to ensure a sufficient transmission for radiography. Due to the low thickness of the specimen, the zinc vapor may exit through the lap joint instead of the processing zone, so a relatively high surface mass (275 g/m^2 , layer thickness approx. $20 \mu\text{m}$ according to EN 10346) was chosen to ensure that effects could occur during welding. It should be noted that penetration welds were performed to except an effect of the AISI 304 sample carrier on the process.

The laser beam was aligned to the specimen's center at $t/2$. The movement of the sample is shown by v_m while v_w represents the resulting welding direction. The procedure ensured that keyhole and melt pool were recorded throughout the welding process at the same position. Compressed air was used as process gas in selected trials to reduce the effect of zinc vapor interacting with the laser beam.

Image preprocessing was carried out with Photron Fastcam Viewer 4 software in the first step. The High Dynamic Range (HDR) option of the software enabled the gray value differences to be increased between keyhole and melt pool and surrounding areas. The calculation of a median image I_{med} consisting of 500 preprocessed frames with the sample moving without laser welding process was calculated in the next step. Each preprocessed frame I_n , with n indexing the frame number, was divided by the median image resulting in the presented image $I_{proc,n}$ as shown in Eq. (1). This procedure allows the reduction of artifacts originating from scintillator, beam path or sample. Adjustments of brightness and contrast ensured a comparable depiction of high-speed images.

$$I_{proc,n} = \frac{I_n}{I_{med}} \quad (1)$$

3. Results and discussion

3.1. General behavior of the keyhole during welding

The keyhole behavior during welding zinc-coated steels is described based on the image series shown in Fig. 2a. The image series starts at t_0 during the piercing of the keyhole, i.e., close to the general start of the laser process. The keyhole, the melt wave above the sheet surface and the zinc-coated interface between both sheets were clearly detectable. It should be noted that the melt pool contour inside the sheet could not be extracted due to the material properties and limited phase contrast between melt and base material.

With increasing times, the keyhole reached the zinc coating, resulting in the formation of a bulge ($t_0+0.575$ ms), followed by a significant upward movement of the melt pool due to the pressure increase originating from zinc evaporation ($t_0+0.65$ ms). Subsequently, melt was ejected from the processing zone ($t_0+0.725$ ms). During these times, further piercing of the keyhole was delayed by approx. 0.4 ms before penetration depth increased again, as shown for $t_0+1.80$ ms. The zinc vapor was leading to additional instabilities, e.g., the formation of melt columns and resulting spatter detachment in the interface region ($t_0+3.575...3.65$ ms). With increasing processing times, a large void area without melt was formed, but melt bridges originated from the flow around the keyhole aperture in the surface region were shadowing the ongoing

phenomena ($t_0+18.9$ ms). A further increase in time showed that the vapor pressure of zinc was interacting with the melt bridges. On the one hand, leading to strong fluctuations and melt ejections. On the other hand, constantly pushing the melt upwards against gravity and opening the melt bridges ($t_0+50.95$ ms). The results in hump-shaped structures that were shown in Fig. 2b for corresponding top view images. In addition, melt bridges broke up and formed droplets laterally attached to the keyhole. When no large melt ejection occurred, the hump-shaped structures were starting to solidify, enclosing a large void region trapped inside the weld ($t_0+53.95$ ms).

The melt pool dynamics were highly volatile, but the keyhole depth was also significantly changing over time. The reason is the interaction of the zinc vapor with the laser beam. By using compressed air as the process gas (transverse flow over the surface of the sheet), it was possible to observe the effect on the keyhole depth based on zinc vapor interacting with the laser beam. Fig. 3 shows the keyhole depth over time for an interval of 50 ms. The sheet surface is represented at 0 while the interface between both sheets was present at a nominal depth of -1.25 mm. The keyhole depth is strongly fluctuating for the initial process and ranges between approx. -1.9 to -1.4 mm, i.e., differences from 500 μ m were observed (see Fig. 3a). In contrast, keyhole depths significantly stabilized by using the process gas ranging between -1.8 and -2.0 mm (Fig. 3b). The keyhole depth reached a quasi-stationary state when compressed air was used as the process gas.

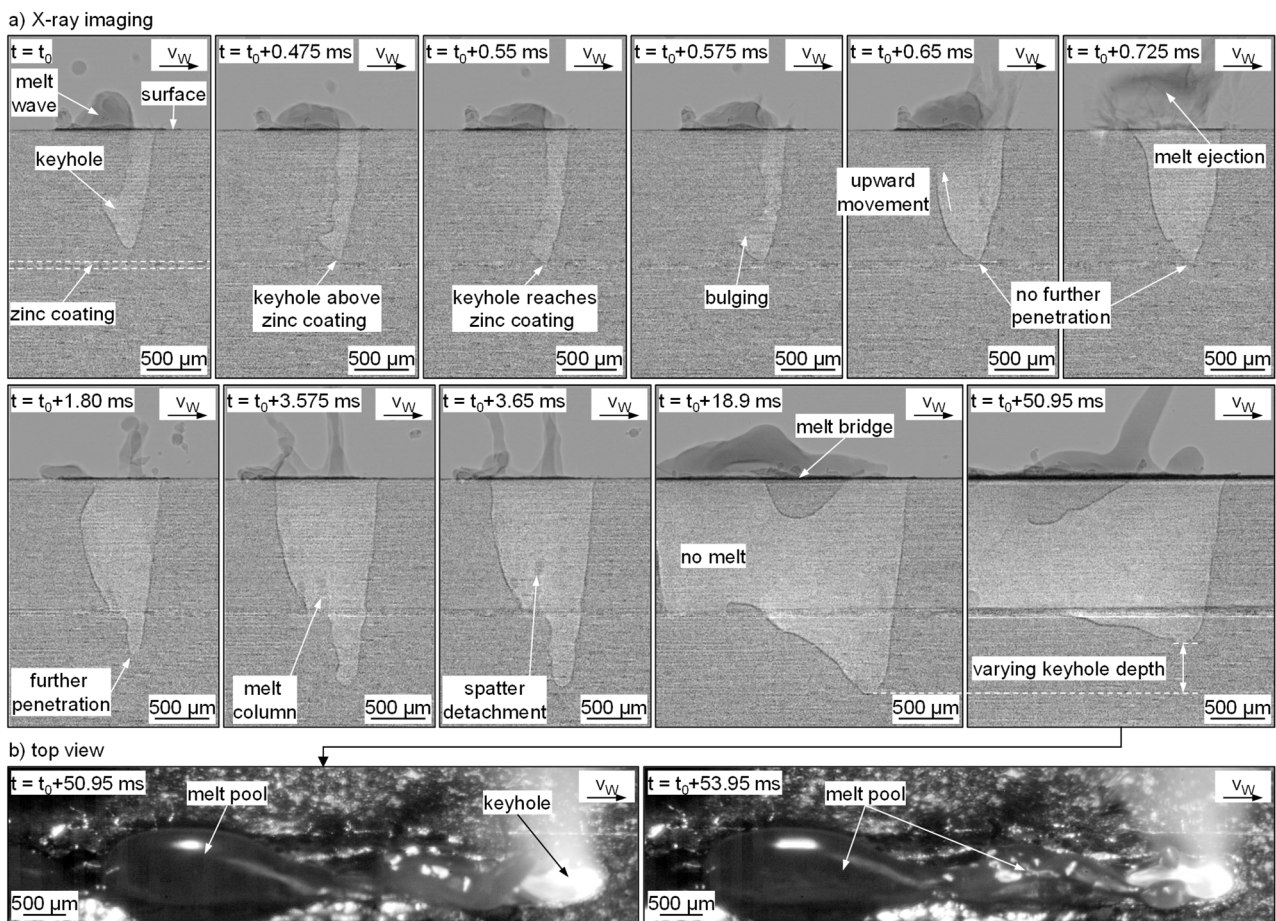


Fig. 2: a) Image series starting from keyhole piercing and b) corresponding top view images ($v_w = 6$ m/min, $P_{L,ring} = 1,750$ W, $P_{L,core} = 250$ W, $z_0 = 0$ mm, no process gas, X-ray acquisition rate f_i)

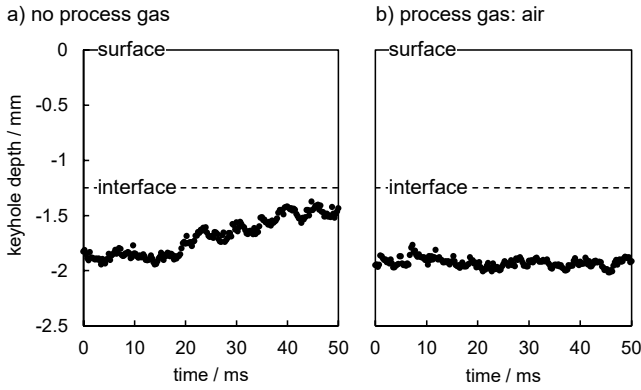


Fig. 3. Keyhole depth over time: a) no process gas; b) with process gas ($v_w = 6$ m/min, $P_{L,ring} = 1,750$ W, $P_{L,core} = 250$ W, $z_0 = 0$ mm, X-ray acquisition rate f_i)

In addition to the changing keyhole depth, the behavior of the keyhole front was studied in more detail, i.e., to describe if disturbances were induced by zinc vapor escaping through the keyhole front.

3.2. Effect of zinc vaporization on keyhole front

The time-dependent behavior of the keyhole front was studied at different depths along the keyhole front. In order to obtain time-dependent information at different positions, streak images were created, i.e., sections with a height of 1 px were cut from the image and arranged in ascending order of time. The different gray values represent the material leading to the keyhole, the keyhole front (reduced gray values led to darker appearance based on phase contrast), and the inner keyhole region.

Fig. 4 shows streak images of the keyhole front for different depths, whereby $z = 0$ represents top of the sheet. Five different positions were considered (indicated by black lines), representing the center region of the top sheet (-0.75 mm), the interface (-1.25 mm) and two adjacent positions in top and bottom sheet ($-1.15/-1.35$ mm) as well as the center region of the keyhole in the bottom sheet (at -1.65 mm). It should be noted that increasing depths showed an enlarged region of material leading to the keyhole due to the inclination angle of the keyhole front. A more detailed description of the behavior of the keyhole front at different depths is given in the following:

- **-0.75 mm:** Stable keyhole front with minor deviations, i.e., no strong fluctuations were observed.
- **-1.15 mm:** Slightly increased deviations at the keyhole front compared to the center region of the top sheet, i.e., zinc vapor is causing minor disturbances.
- **-1.25 mm:** The interface region showed differences in gray values due to the zinc coating and edge effects which is why the contrast between keyhole front and neighboring areas was reduced. A slight increase in variations of the keyhole front position can be discussed based on the effect of zinc vapor escaping from the zero-gap region, however, all deviations occurring are significantly smaller than $100 \mu\text{m}$.
- **-1.35 mm:** Comparable to -1.15 mm
- **-1.65 mm:** Comparable to -0.75 mm

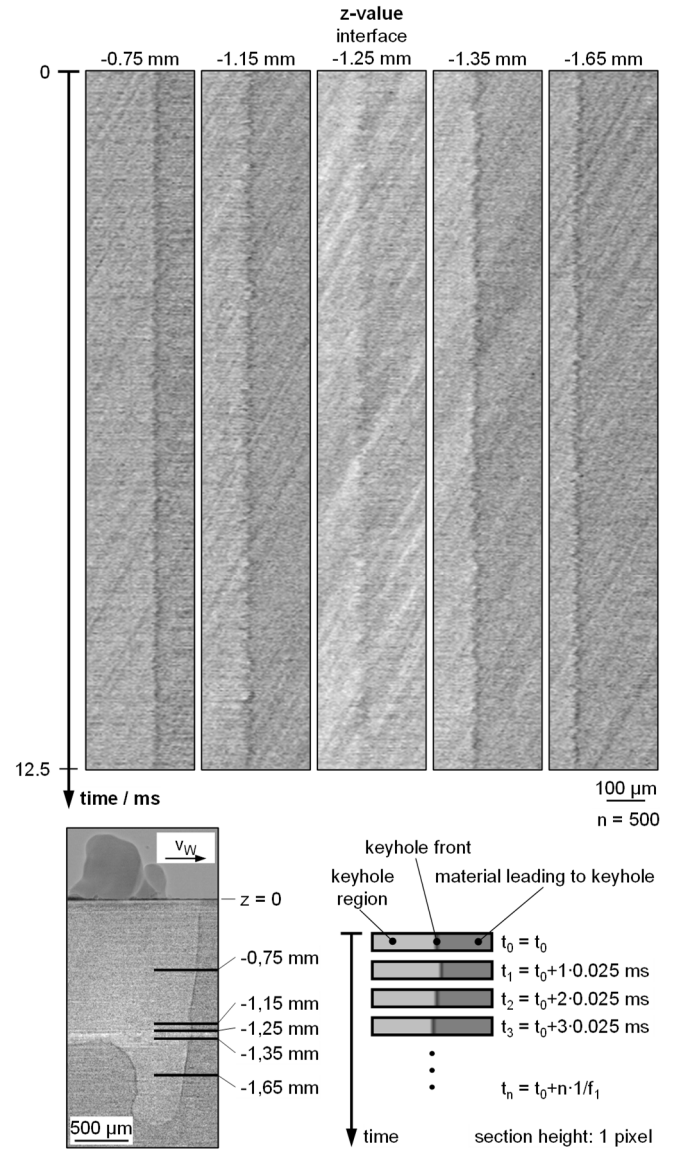


Fig. 4. Streak images of keyhole front at different positions ($v_w = 6$ m/min, $P_{L,ring} = 1,750$ W, $P_{L,core} = 250$ W, $z_0 = 0$ mm, process gas: air, X-ray acquisition rate f_i)

Although the interaction of the zinc vapor with the laser beam affected the keyhole depth (see Fig. 3), significant instabilities at the keyhole front, as described for zero-gap welding in the state of the art [6], were not detected in the quasi-stationary phase of the welding process. To further investigate the effect of zinc vapor on defect formation, its impact on the melt pool following the keyhole was studied.

3.3. Effect of zinc evaporation on melt pool

The field of view of the synchrotron setup was enlarged to obtain more information from the melt pool (acquisition rate f_2 , see section 2.2). In section 3.1 it was shown that the vapor pressure of zinc led to melt bridges formed on the top side of the weld accompanied by large void areas below these bridges. To avoid the continuous formation of such voids, the laser beam was defocused ($z_0 = -3$ mm) and the parameters were adjusted to ensure partial penetration welds.



Fig. 5. Image series on the effect of zinc vaporization on melt pool behavior ($v_W = 6$ m/min, $P_{L,ring} = 2,000$ W, $P_{L,core} = 750$ W, $z_0 = 3$ mm, process gas: air, X-ray acquisition rate f_2)

The effect of zinc vaporization on the melt pool behavior is explained based on the image series shown in Fig. 5. In general, the formation of melt bridges on the top side was avoided and the keyhole region appeared strongly elongated and widely opened towards the sheet surface due to zinc vapor pressure. This zinc vapor is assumed to originate from the regions lateral to the weld, i.e., the widening temperature field led to further evaporation of zinc interacting with the rear melt pool [7].

A trailing melt pool wave indicated the end of the keyhole region (t_0). Minor instabilities occurred at certain points in time where the melt pool crosses both sheets at the interface. Increased evaporation of zinc resulted in a further displacement of melt in this region until degassing in the keyhole region took place ($t_0+0.05\ldots0.30\text{ ms}$) without causing significant imperfections. However, the evaporation of zinc was not constant over time, i.e., deviations in evaporation pressure occurred, accelerating the melt pool wave and lateral melt flowing around the keyhole region in upward direction leading to spatter formation ($t_0+2.95\ldots5.10\text{ ms}$). The vapor pressure prevented the melt pool to crossing the interface between top and bottom sheet again and solidification starts leading to the entrapment of voids ($t_0+5.75\ldots6.45\text{ ms}$). The laterally degassing zinc was still acting at the interface ($t_0+18.50\text{ ms}$) when a significant evaporation event occurred at the rear of the melt pool ($t_0+18.65\text{ ms}$) leading to a fast displacement of the whole melt pool and melt ejection ($t_0+18.90\text{ ms}$). The resulting melt columns are elongated due to the high kinetic energy and formation of spatters ($t_0+19.30\ldots21.10\text{ ms}$). On the one hand, the lateral evaporation of zinc was fast ejecting the whole melt pool. On the other hand, the spatters were shadowing the laser beam, i.e., absorption at the keyhole front was prevented. When the laser beam started interacting with the keyhole front again and penetration began to increase again, additional voids were formed at the interface ($t_0+21.10\ldots22.45\text{ ms}$). It should be noted that when the keyhole starts to cross the interface again, instabilities as seen during the initial piercing process (see section 3.1) occurred again.

4. Conclusion

In this paper, in situ high-speed synchrotron X-ray imaging was conducted during laser beam welding of zinc-coated steels in zero-gap lap joint configuration and provided novel insights into the interaction between zinc vapor with the keyhole and melt pool. The study allowed to determine effects leading to pore and spatter formation as well as ejections of the whole melt pool at a high temporal and spatial resolution. The effect of zinc vaporization during the piercing of the keyhole and quasi-stationary welding process was described, whereby the zinc vapor pressure led to the formation of hump-shaped structures at the surface enclosing a large void area inside the weld. The keyhole depth fluctuated due to zinc vapor interacting with the laser beam. However, the keyhole front was determined to show small disturbances over time by comparing different positions based on streak images. The interaction between zinc vapor and melt pool was demonstrated by increasing the field of view and showing the time-dependent

events leading to melt and spatter ejection and repercussions on the welding process affecting the penetration depth.

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